Official Review of Submission28088 by Reviewer vpXC

Official Reviewby Reviewer vpXC13 Mar 2026 at 04:12 (modified: 24 Mar 2026 at 07:34)Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer vpXCRevisions

Summary:

This paper statistically benchmarks two-way attention transformers for multivariate time-series forecasting in low-data and low-SNR regimes via synthetic data with tunable temporal/cross-sectional dependencies. It shows the model outperforms classic baselines (Lasso, boosting, MLP) in most scenarios (e.g., cross-sectional shifts, non-linearities) but fails at simultaneous time-cross-section shifts. The authors propose a data-driven dynamic attention sparsification method that boosts performance in noisy environments, analyze interpretable sparse attention patterns, and provide an analytical OLS performance derivation to link empirical results with statistical theory.

Strengths And Weaknesses:

Strengths:

(1) The authors applied bidirectional attention to time series forecasting scenarios and, by constructing controllable datasets, clarified its advantages and limitations in low-data and low signal-to-noise ratio (SNR) settings.
(2) They verified that dynamic sparse attention mechanisms can improve performance in high-noise environments without degrading performance in scenarios without sparsity.
(3) The conclusions of the paper provide insights for future research directions.

Weaknesses:

(1) This paper lacks novelty; it appears to simply test existing methods in low-data, high-noise scenarios and report the results and conclusions. Although the paper highlights the benefit of achieving sparsity by retaining only the top K% of attention scores, this has already been pointed out in LSINet[1].
[1] A lightweight sparse interaction network for time series forecasting. AAAI
(2) The paper only compares with simple traditional methods and lacks comparisons with more modern deep learning transformer and MLP models for time series forecasting, such as TimePro[2], SEMixer[3], TFPS[4], to demonstrate the SOTA of the paper's method.
[2] TimePro: Efficient Multivariate Long-term Time Series Forecasting with Variable- and Time-Aware Hyper-state.
[3] SEMixer: Semantics Enhanced MLP-Mixer for Multiscale Mixing and Long-term Time Series Forecasting.
[4] Learning Pattern-Specific Experts for Time Series Forecasting Under Patch-level Distribution Shift.
(3) The literature review is insufficient, and both readability and writing quality need further improvement.
(4) The experimental details are not described adequately, making the results difficult to reproduce.

Soundness: 2: fair
Presentation: 2: fair
Significance: 2: fair
Originality: 2: fair

Key Questions For Authors:

Please refer to the Strengths and Weaknesses.

Limitations:

The authors should conduct a more comprehensive literature review and include experiments with additional Transformer- and MLP-based forecasting models to broaden and strengthen the persuasiveness of their conclusions.

Overall Recommendation: 2: Reject: For instance, a paper with technical flaws, weak evaluation, inadequate reproducibility, incompletely addressed ethical considerations, or writing so poor that it is not possible to understand its key claims.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Compliance With LLM Reviewing Policy: Affirmed.

Code Of Conduct Acknowledgement: Affirmed.

Official Review of Submission28088 by Reviewer bPFr

Official Reviewby Reviewer bPFr11 Mar 2026 at 10:03 (modified: 24 Mar 2026 at 07:34)Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer bPFrRevisions

Summary:

This work explores when transformer architectures are effective in multivariate time-series forecasting under low signal-to-noise ratio and low sample size. Instead of improving scores on real benchmarks, the authors construct a controllable synthetic statistical benchmark. They generate data with known dependency structures and decompose the prediction target into several canonical effects. Then they systematically vary the signal-to-noise ratio and repeat experiments using bootstrap to evaluate how different models correlate with the optimal ground-truth predictor. In addition, the authors propose a dynamic attention sparsification method.

Strengths And Weaknesses:

Strengths

1. The work tries to answer a research question that is more meaningful than simply achieving higher scores on benchmarks: when and under what structures transformer models have advantages.
2. By controlling the data generation process and separating different types of dependencies, the authors can analyze mechanisms more clearly than simply stacking experiments on multiple real datasets. In particular, they use the known ground-truth predictor to evaluate out-of-sample correlation, which is closer to testing whether the model learns the correct structure than simply reporting MSE or MAE.
3. The experiments are not run only once but repeated using bootstrap, and significance tests are performed for full attention and max-sparse attention. Although not complete, it is still more careful than many works that only report average results.

Weaknesses

1. Very weak external validity because the study almost completely relies on synthetic data.The entire paper is almost fully based on synthetic data generated from the authors' designed DGP. If the conclusions only hold on the canonical effects defined by the authors, then the results are more like evaluating how models perform on a test designed by the authors rather than providing reliable insights into real forecasting problems. At least the idea should be tested on some real benchmarks and compared with other models to show its effectiveness.
2. Baseline selection is insufficient and may favor transformer models. The paper mainly compares Lasso, Boosting, and fully connected MLP. This set of baselines is not strong enough, especially for time-series forecasting. There are many recent SOTA methods. Including foundation models such as MOMENT [1] and Time-MoE [2], more complex models such as TimeMixer++ [3], PatchTST [4], or iTransformer [5], and also lightweight fast models such as DLinear [6] and SparseTSF [7] would strengthen the evaluation.
3. The proposed dynamic sparse attention method is shallow and lacks novelty. The theoretical support is weak. The authors only provide an intuitive example and do not explain under what conditions this rule holds. It is also unclear what is fundamentally different from existing sparse attention or cross-attention approaches.
4. The work shows that transformer performs poorly on TSCS-Shift while MLP performs the best. This result is important because it shows that the proposed architecture struggles with joint temporal and cross-sectional shift dependencies. However, the paper does not provide deeper analysis of this phenomenon.
5. The evaluation metric is limited and disconnected from common forecasting metrics. The authors design their own metric, which is acceptable, but the evaluation should also include common forecasting metrics instead of relying only on the proposed metric.
6. Many conclusions are only explained verbally and lack systematic ablation studies. The design choices of the transformer architecture are not sufficiently justified. The sparsification threshold K=0.1K=0.1K=0.1 seems arbitrarily chosen without hyperparameter tuning or sensitivity analysis. There is no discussion of computational cost, training stability, or hyperparameter sensitivity. Although the title suggests a benchmark, the work is closer to a case study within a synthetic framework because it does not include many baselines from the last five years.

[1] Goswami M, Szafer K, Choudhry A, et al. MOMENT: A Family of Open Time-series Foundation Models[C]//Forty-first International Conference on Machine Learning.

[2] Shi X, Wang S, Nie Y, et al. Time-MoE: Billion-Scale Time Series Foundation Models with Mixture of Experts[C]//The Thirteenth International Conference on Learning Representations.
[3] Wang S, Jiawei L I, Shi X, et al. TimeMixer++: A General Time Series Pattern Machine for Universal Predictive Analysis[C]//The Thirteenth International Conference on Learning Representations.
[4] Nie Y, Nguyen N H, Sinthong P, et al. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers[C]//The Eleventh International Conference on Learning Representations.
[5] Liu Y, Hu T, Zhang H, et al. iTransformer: Inverted Transformers Are Effective for Time Series Forecasting[C]//The Twelfth International Conference on Learning Representations.
[6] Zeng A, Chen M, Zhang L, et al. Are transformers effective for time series forecasting?[C]//Proceedings of the AAAI conference on artificial intelligence. 2023, 37(9): 11121-11128.
[7] Lin S, Lin W, Wu W, et al. SparseTSF: Modeling Long-term Time Series Forecasting with* 1k* Parameters[C]//Forty-first International Conference on Machine Learning.

Soundness: 2: fair

Presentation: 1: poor

Significance: 1: poor

Originality: 2: fair

Key Questions For Authors:

refer to the weaknesses

Limitations:

refer to the weaknesses

Overall Recommendation: 1: Strong Reject: For instance, a paper with well-known results, unaddressed ethical considerations, or a poorly written paper where it is impossible to tell what the nature of its contribution is.

Confidence: 5: You are absolutely certain about your assessment. You are very familiar with the related work and checked the math/other details carefully.

Compliance With LLM Reviewing Policy: Affirmed.

Code Of Conduct Acknowledgement: Affirmed.

Official Review of Submission28088 by Reviewer PawB

Official Reviewby Reviewer PawB09 Mar 2026 at 15:35 (modified: 24 Mar 2026 at 07:34)Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer PawBRevisions

Summary:

The paper proposes a benchmark for studying transformer architectures in multivariate time-series forecasting under varying signal-to-noise ratios. Using synthetically generated data with known dependency structures, the authors evaluate a transformer that alternates temporal and cross-sectional self-attention blocks against classical baselines. The paper also introduces a dynamic sparsification mechanism for attention matrices and provides an analytical derivation of expected OLS performance. The results suggest that the transformer's advantage is concentrated on non-linear feature interactions and cross-sectional shifts, while Lasso remains the stronger model for linear and temporal-shift dependencies.

Strengths And Weaknesses:

Strengths:

- The use of synthetic data with known ground-truth dependencies to study transformer behavior in a controlled setting is appealing, and I think there is genuine value in this kind of analysis for the community.
- The finding that the transformer is the only model capable of detecting non-linear feature interactions provides a concrete example of where attention mechanisms add value over classical approaches. The appendices also add more theoretic insight to the findings in the main sections.
- The bootstrapped experimental design, where datasets are regenerated to assess statistical significance of performance differences, demonstrates rigor in the experiments.

Weaknesses:

- The paper's front page deviates from the standard ICML template, which is a submission requirement.
- The temperature forecasting example provided in the paper seems to be at odds with the actual data generation process, which

samples features from a standard Gaussian with a time-invariant mapping to the target. This bears little resemblance to real temperature data where seasonality, autocorrelation, and spatial correlations are fundamental. I found it hard overall to assess the practical relevance of the findings.
- While I appreciate the controlled synthetic setup, the paper contains no real-world data, and I think even a single low signal-to-noise dataset as a sanity check would have gone a long way. The stationarity and Gaussian assumptions soften the conditions that make time-series forecasting challenging in practice, so it is unclear whether the findings would transfer to more realistic settings.
- The introduction cites several transformer architectures and the work questioning transformer effectiveness, but I don't think these appear in the experiments? I appreciate that the paper's focus is on understanding the proposed attention mechanism rather than running an exhaustive benchmark, though testing only one custom transformer design may limit the utility of the findings to practitioners.
- The paper contains no figures at all in the main text, with all results presented as numerical tables. This made it unnecessarily difficult to grasp the trends across noise levels and dependency types. Even plots of the synthetic time series themselves would help build intuition for the different effects and improve readability. Taken together with the template issue, these presentation concerns left me feeling the paper is not quite at the expected publication standard.

I ultimately found it a bit difficult to identify a clear takeaway from the paper. The controlled synthetic framework is a reasonable idea that I think could work well in a more developed form, but the bridge from these synthetic findings to actionable insight for the broader community didn't land for me.

Soundness: 3: good
Presentation: 1: poor
Significance: 2: fair
Originality: 2: fair
Key Questions For Authors:
- Would it be helpful to include at least one real-world low signal-to-noise dataset to provide some evidence that the synthetic results transfer beyond the paper settings?
- Several transformer architectures cited in the introduction are not benchmarked. Was this a deliberate scope decision and would you expect the findings to generalize to these architectures?

Limitations:
Nothing major, although clearer implications of the controlled assumptions made in this experimental set-up on practical real-world utility would be helpful.
Overall Recommendation: 3: Weak reject: A paper with clear merits, but also some weaknesses, which overall outweigh the merits. Papers in this category require revisions before they can be meaningfully built upon by others. Please use sparingly.
Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
Compliance With LLM Reviewing Policy: Affirmed.
Code Of Conduct Acknowledgement: Affirmed.

Official Review of Submission28088 by Reviewer iutV

Official Reviewby Reviewer iutV04 Mar 2026 at 14:37 (modified: 24 Mar 2026 at 07:34)Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer iutVRevisions

Summary:
The paper studies the capabilities of transformer architectures for multivariate time-series forecasting in environments with limited data and significant low SNR. Instead of standard real-world benchmarks where the underlying data-generating processes are unknown, the authors construct a controlled synthetic framework with tunable structural dependencies. A factorized two-way attention transformer is evaluated against flattened global baselines (e.g., OLS, Lasso, Boosting, MLP). Additionally, the authors propose max_sparse, a data-driven dynamic sparsification algorithm. Overall, this is an insightful synthetic study, but also limited external validity without more realistic processes or real-data confirmation.

Strengths And Weaknesses:

Strength:

1. **Originality:** Evaluating architectures on synthetic datasets with known DGPs is a commendable methodology since it isolates specific spatial, temporal, and non-linear inductive biases, and allows for a mechanistic understanding of what these models can actually learn and whether transformers genuinely learn spatio-temporal dynamics.

2. **Significance:** The paper focuses on Low-SNR setting, which is a well-known, notorious issue

in domains like quantitative finance or biomedicine. Testing models at extreme noise levels explicitly addresses a practical gap in the deep learning time-series community.

3. **Clarity and Honest Reporting:** It is good to see the transparent reporting of negative results, such as the transformer completely failing on the TSCS-Shift and the Lasso outperforming it on the superposition of all effects, which builds credibility of this paper.
4. The manuscript is clearly structured and easy to follow.

Weakness:

1. **(Major)The synthetic data is too idealized.** By generating the input features as strictly independent standard Gaussians across time, series, and features (Sec 2.3), the authors have erased the core properties of real-world time-series (e.g., autocorrelation, trends, seasonal, etc.). The problem they constructed is somehow equivalent to a spatial feature-selection task on a 3D grid with static noise. I believe claiming that these models "generalize effectively" in time-series forecasting is overstated given this purely i.i.d. design.
2. **(Major) Empirical finding undermines the core structural claim.** Table 5 shows the proposed Transformer (Trans) is "collapsing" on the TSCS-Shift effect, achieving near-zero correlations, while the simple MLP performs substantially better (at). The authors openly admit this and leave this problem for a future work. While I appreciate the authors' honesty and transparency, the entire premise of proposing a "Two-Way" (Temporal and Cross-Sectional) attention mechanism is to capture complex spatio-temporal dynamics. The fact that the model completely fails on the only canonical dependency that mixes time and cross-section undermines and contradicts the core architectural design of the paper.
3. **(Technical) Differentiable and possibly dead attention.** The dynamic sparsification relies on a hard threshold. Setting logits to is a non-differentiable step function and in autodiff frameworks (like PyTorch), applying a softmax over yields , entirely cut-off the gradient flow to those masked logits during backpropagation. The network receives no mathematical feedback on why it made a masking decision or how to adjust its representations to get a better mask next time.
4. Baselines in this paper are relatively narrow and somehow unfavorable. The MLP baseline is a large fully-connected network on a flattened vector, no structured baselines like TCN[1], RNN[2], N-BEATS[3], simple linear TS models (e.g., AR), or modern "linear" TS baselines discussed in the intro are included.

[1]Lea, Colin, et al. "Temporal convolutional networks: A unified approach to action segmentation." European conference on computer vision. Cham: Springer International Publishing, 2016.
[2]Sherstinsky, Alex. "Fundamentals of recurrent neural network (RNN) and long short-term memory (LSTM) network." Physica d: Nonlinear phenomena 404 (2020): 132306.
[3]Oreshkin, Boris N., et al. "N-BEATS: Neural basis expansion analysis for interpretable time series forecasting." arXiv preprint arXiv:1905.10437 (2019).

Soundness: 2: fair
Presentation: 2: fair
Significance: 2: fair
Originality: 2: fair
Key Questions For Authors:

1. Dynamic sparsity implementation: Is the mask treated as stop-gradient? Is implemented as a large negative constant?
2. Does the batch-averaging of attention create instability or dependence on batch composition?

Limitations:
yes

Overall Recommendation: 3: Weak reject: A paper with clear merits, but also some weaknesses, which overall outweigh the merits. Papers in this category require revisions before they can be meaningfully built upon by others. Please use sparingly.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Compliance With LLM Reviewing Policy: Affirmed.

Code Of Conduct Acknowledgement: Affirmed.